5.5.3 Cosmology

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	distances measured in astronomical unit (AU), light-year (ly) and parsec (pc)	*M4.6*
(b)	stellar parallax	
(c)	the equation $p = \frac{1}{d}$, where p is the parallax in seconds of arc and d is the distance in parsec	
(d)	the Cosmological principle; universe is homogeneous, isotropic and the laws of physics are universal	
(e)	Doppler effect; Doppler shift of electromagnetic radiation	
(f)	Doppler equation $\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$ for a source of electromagnetic radiation moving relative to an observer	
(g)	Hubble's law; $v \approx H_0 d$ for receding galaxies, where H_0 is the Hubble constant	HSW7
(h)	model of an expanding universe supported by galactic red shift	HSW2, 7, 8, 11
(i)	Hubble constant H_0 in both $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $\mathrm{s^{-1}}$ units	
(j)	the Big Bang theory	HSW7, 9, 10, 12

(k)	experimental evidence for the Big Bang theory from microwave background radiation at a temperature of 2.7 K	HSW7, HSW11 The development and acceptance of Big Bang theory by the scientific community.
(l)	the idea that the Big Bang gave rise to the expansion of space-time	
(m)	estimation for the age of the universe; $t \approx H_0^{-1}$	*M1.4* HSW7
(n)	evolution of the universe after the Big Bang to the present	HSW1, 2, 5, 6, 7, 8, 9, 10, 11
(o)	current ideas; universe is made up of dark energy, dark matter, and a small percentage of ordinary matter.	